

Words Beyond the Rate: High-Frequency Monetary Policy Shocks and FOMC Language

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Outline

- 1 Motivation
- 2 Data & Methodology
- 3 Results
- 4 Robustness
- 5 Conclusion

FOMC Language Moves Markets — Not Just Rates

The Taper Tantrum (2013)

- FOMC kept rates unchanged
- Language shift: “may reduce” asset purchases
- 10Y Treasury yield rose 40 bp in days
- Markets reacted to **words**, not the rate

Core Question

Does FOMC statement language convey information **beyond** the rate decision itself?

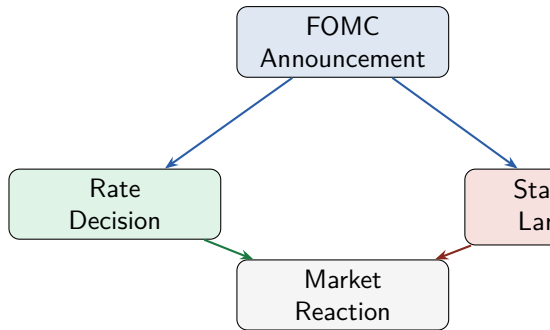


Figure 1. Two channels of FOMC influence

Two Competing Interpretations

Policy Implementation

- Statement **explains** the decision
- Target shock drives sentiment
- Language reflects *what was done*
- Bernanke & Kuttner (2005)

Informational Revelation

- Statement **reveals** future policy intent
- Path shock (FG) drives sentiment
- Language signals *what will be done*
- Romer & Romer (2000), Campbell et al. (2012)

Our contribution: Decompose which channel dominates using GSS high-frequency shocks + textual sentiment analysis

Data Overview: 117 FOMC Meetings (2006–2022)

Monetary Policy Shocks

- **Target:** GSS current-rate surprise
- **Path:** GSS future-policy revision
- Source: Acosta (2022)
- $\rho = 0.14$ (successfully separated)

Sentiment Measures

- LM: Loughran-McDonald
- CB: Central-bank-specific
- Combined: $0.5 \times \text{LM} + 0.5 \times \text{CB}$

Financial Market Data

- CRSP VW/EW/S&P 500 (WRDS)
- NASDAQ, Gold
- 10Y Treasury, 13W T-bill
- VIX index

Key Features

- 3 Chairs: Bernanke, Yellen, Powell
- 3 Regimes: conventional, FG, normalization
- FG: Dec 2008 – Dec 2015 (57 meetings)

Empirical Framework

H1: Sentiment $\sim$ Shocks

$$S_t = \alpha + \beta_T \cdot \text{Target}_t + \beta_P \cdot \text{Path}_t + \varepsilon_t$$

H2: Returns $\sim$ Shocks

$$R_t = \alpha + \beta_T \cdot \text{Target}_t + \beta_P \cdot \text{Path}_t + \varepsilon_t$$

H3: Wald Test ($\beta_T = \beta_P?$)

$$W = \frac{(\hat{\beta}_T - \hat{\beta}_P)^2}{\text{Var}(\hat{\beta}_T - \hat{\beta}_P)} \sim \chi^2(1)$$

H4: FG Interaction

$$R_t = \alpha + \beta_T T_t + \beta_P P_t + \beta_3 S_t + \beta_4 (S_t \times \text{FG}_t) + \varepsilon_t$$

All regressions: OLS with **Newey-West HAC(4)** standard errors.

Target Shock Significantly Predicts Statement Sentiment

Table 1. Sentiment and Monetary Policy Shocks (H1)

Variable	β	SE	t	p
Target shock	0.000577	0.000238	2.43	0.017
Path shock	0.000633	0.000439	1.44	0.152
Constant	0.0145	—	—	—
R^2	1.57%			
N	117			

Key Finding

Target shock is **significant** ($p = 0.017$); path shock is **not** ($p = 0.152$)

Caveat

$R^2 = 1.57\%$ is modest — typical for text-based event-study regressions

Proper Shock Identification Matters

Table 2. Surprise Measure Comparison

Surprise Measure	$\beta(t)$	p	R^2	N
Rate change	0.00190 (0.64)	0.525	0.40%	117
Kuttner surprise (bp)	0.00023 (2.61)	0.009	1.49%	117
GSS target + path	0.00058 (2.43)	0.017	1.57%	117

Implication

Simple rate changes **cannot** detect the sentiment–policy link ($p = 0.525$). High-frequency identification is essential for communication studies.

Target Shocks Drive Cross-Asset Responses

Table 3. Asset Returns and Monetary Policy Shocks (H2)

Asset	β_T	p_T	β_P	p_P	R^2	N
CRSP VW	-0.435	0.043	-0.186	0.443	9.10%	117
CRSP EW	-0.449	0.013	-0.174	0.479	10.30%	117
S&P 500	-0.391	0.073	-0.179	0.424	7.80%	117
NASDAQ	-0.282	0.039	-0.166	0.309	3.40%	117
Gold	-0.404	0.014	-0.488	0.146	7.00%	117
10Y Treasury	0.007	0.403	-0.001	0.890	0.70%	117
13W T-bill	0.004	0.491	-0.003	0.737	0.70%	117

EW > VW > S&P: smaller firms more sensitive $\Rightarrow$ financing-channel evidence

Evidence Favors Policy Implementation Over Information Revelation

H3: Wald Test — $\beta_T = \beta_P$?

- $\chi^2 = 0.015$
- $p = 0.902$
- **Cannot reject** equality
- Coefficients are statistically indistinguishable

Interpretation

While target shock is individually significant and path is not, we **cannot formally distinguish** their magnitudes. Evidence *favors* implementation but does not eliminate revelation.

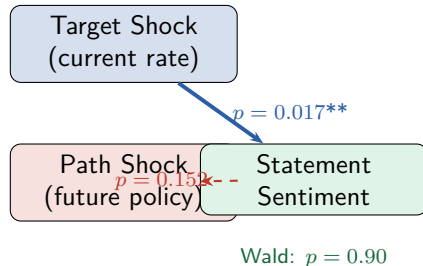


Figure 2. Implementation vs. Revelation

Forward Guidance Does Not Strengthen the Language Channel

Table 4. Forward Guidance Period Interaction (H4)

Variable	CRSP VW	NASDAQ
Target shock	−0.421 (0.046)	−0.289 (0.031)
Path shock	−0.175 (0.452)	−0.166 (0.320)
Sentiment	−20.73 (0.191)	5.47 (0.709)
Sentiment × FG	−3.72 (0.836)	6.04 (0.739)
R^2	9.9%	3.5%

Result

Interaction not significant – sentiment does not become more important during FG period

Results Robust Across Sentiment Measures and Documents

Table 5. Alternative Sentiment Measures

Model	β_T	p_T	β_P	p_P	R^2
Statement Comb.	0.000577**	0.017	0.000633	0.152	1.57%
Minutes LM	0.000918*	0.083	0.001091	0.324	3.67%
Minutes CB	0.000147	0.716	0.001423*	0.061	5.60%
Minutes Comb.	0.000532**	0.011	0.001257**	0.015	9.35%
Statement + Min.	0.000391*	0.062	0.000194	0.611	6.06%

CB Outperforms LM

CB captures hawkish/dovish language that LM misses

Minutes Reveal Path Info

Path shock significant ($p = 0.015$) in Minutes

Robustness: Standard Errors, COVID, Regimes

Alternative SE & Lags

- NW HAC(2), HAC(4), HAC(6)
- White heteroskedasticity-robust
- Target significant across all
- Path insignificant across all

Excluding COVID

- Results unchanged
- Target: $p \approx 0.02$
- Path: $p \approx 0.16$

Regime Heterogeneity

- **FG** ($N=61$): Target $p < 0.001$
- **Norm.** ($N=49$): Neither significant
- **Conv.** ($N=7$): Too few obs.

Bauer-Swanson (2023)

Shocks partially predictable ($R^2 \approx 11\text{--}14\%$), but critique applies to *both* target and path

Extension: JK Sign-Restriction Decomposition

Method (Jarociński & Karadi 2020)

Classify target shocks by sign restriction:

- **MP shock**: rate $\uparrow$, stocks $\downarrow$ (59%)
- **CBI shock**: rate $\uparrow$, stocks $\uparrow$ (41%)

CBI = central bank information effect

H1: Sentiment Regression

	Coeff.	p
β_{MP}	0.000541	0.134
β_{CBI}	0.000653	0.170
β_P	0.000643	0.159
R^2	1.58%	

Key Finding

Neither MP nor CBI component significant for sentiment; F-test cannot reject $\beta_{MP} = \beta_{CBI}$ ($p = 0.87$). Decomposition loses significance due to reduced power in subgroups.

Extension: JK Decomposition — Asset Returns

H2: CRSP VW Returns

	Coeff.	<i>p</i>
β_{MP}	-1.029	< 0.001***
β_{CBI}	0.844	< 0.001***
β_P	-0.019	0.911
R^2	35.69%	

Interpretation

- **MP shocks:** contractionary → stocks fall ($\beta = -1.03^{***}$)
- **CBI shocks:** rate hike + stocks rise → information effect ($\beta = +0.84^{***}$)
- **Path shock:** still insignificant
- R^2 jumps from 9.1% to 35.7%

JK decomposition **confirms** the information effect in asset markets, but sentiment does not differentiate MP from CBI

Extension: Bauer-Swanson Orthogonalization

Method

Regress shocks on pre-FOMC macro info, use residuals:

- Target predictability: $R^2 = 10.5\%$
- Path predictability: $R^2 = 13.8\%$

Controls: lagged returns, VIX, term spread, rate changes

H1: Sentiment (Orthogonalized)

	Original	Orthogonalized
β_T	0.000592** (0.012)	0.000631 (0.108)
β_P	0.000666 (0.131)	-0.000963 (0.193)
R^2	1.70%	2.00%

Key Finding

Target loses significance ($p = 0.012 \rightarrow 0.108$) after B-S orthogonalization, but path remains insignificant. For asset returns, target **strengthens** ($p = 0.043 \rightarrow 0.005$).

New Finding: Sentiment Has Incremental Power in FG Period

Full Sample (H2)

	$\hat{\beta}_S$	p	Weak in-
CRSP VW	-0.199	0.185	
CRSP EW	-0.101	0.566	
S&P 500	19.46	0.088*	
10Y Treasury	1.24	0.098*	
Incremental power overall			

FG Period: Sentiment Dominates

	FG ($N=57$)	Non-FG ($N=60$)
$\hat{\beta}_S$ (VW)	-2.60*** (0.004)	-0.05 (0.727)
R^2	30.6%	5.6%
$\hat{\beta}_S$ (EW)	-2.70*** (0.009)	0.04 (0.815)
R^2	31.8%	7.7%

Robustness

Permutation test $p = 0.000$; Leave-one-out max $|\Delta\hat{\beta}| = 0.54$; Partial corr (FG) = -0.364 ($p = 0.005$)

Why FG Period? Language as Primary Transmission Channel

The Mechanism

- ZLB: rate tool constrained $\rightarrow$ language becomes the signal
- FG period: FOMC *must* communicate through words
- Non-FG: rate action speaks for itself

Not a Power Issue

FG sentiment $\sigma = 0.0019$

Non-FG sentiment $\sigma = 0.0078$

$\rightarrow 4\times$ **less** variation, but **more** predictive

Reframes the FG Null Result

- Old: “FG doesn’t strengthen path shock” (TRUE but misses the point)
- New: “FG strengthens **sentiment channel**” (TRUE and highly significant)
- Interaction: $\hat{\beta}_{S \times FG} = -2.87^{***}$
($p = 0.005$)

Key Insight

Full-sample weakness = averaging two regimes: language redundant (normal) vs. language dominant (ZLB)

Summary of Findings

What We Find

- 1 Target shock **significantly** predicts statement sentiment ($p = 0.017$)
- 2 Path shock does **not** ($p = 0.152$)
- 3 Target shock drives equity responses; path effects are weak
- 4 $EW > VW$: financing-channel evidence
- 5 **NEW**: Sentiment has **incremental** power in FG period ($R^2 = 30.6\%^{***}$)

What It Means

- Language reflects **implementation** in normal times
- Language becomes **primary channel** at ZLB
- Path shock $\neq$ information shock
- Different documents serve different roles

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Relation to Fernández-Fuertes (2025)

FF: Better Shock Measure

- Multi-agent LLM → narrative surprises
- Processes Statements, Minutes, Beige Books, Press Conferences
- His Table 32: target $\beta_T = 0.047^{***}$, path $\beta_P \approx 0$
- **Same target-dominant pattern**
- $R^2 = 12.4\%$ vs. our 1.57%

Us: What Does Language Convey?

- Different question, not different method
- 4 structured hypotheses + Wald + FG interaction
- Statement vs. Minutes channel comparison
- Fully transparent & reproducible
- No API dependencies

FF's Table 32 is **independent confirmation** of our core finding — complementary, not competing

Future Directions

Methodological Extensions

- LLM-based sentiment (Chen et al. 2025; FF 2025)
- Bauer-Swanson orthogonalized shocks
- Intraday event windows (TAQ data)
- JK (2020) info vs. monetary decomposition

Data Extensions

- Extend GSS series to 2025 (USMPD)
- Press conference transcripts
- SEP projections and dot plots
- Cross-country central bank communication

Key insight: FOMC language is informative, but monetary policy shocks are just **one** of many forces shaping

Thank You

Questions & Discussion

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